UDYAM360 — PRIVACY, SECURITY & DATA PROTECTION NOTE

Airtel SYNC Hackathon 2026 • Enterprise Track A • Technical Safeguards & Prototype Boundaries

1. Purpose of Data Collection: Udyam360 collects minimal enterprise and operational data (business type, employee count, location, telecom spend, and GSTIN/Udyam certificates) strictly for determining telecom product eligibility, configuring accurate commercial quotes, and automating regulatory onboarding verification.

2. Explicit DPDP Consent Process
An unskippable digital consent gate precedes all KYC document operations. Users are informed of specific verification purposes and must check a mandatory acknowledgement. Digital timestamps and consent IDs are recorded in the verification audit trail.

3. Local OCR Processing (Zero Cloud Egress)
Document text extraction is executed 100% locally using an on-premises RapidOCR (ONNX runtime) and WinOCR engine. Scanned certificate files are parsed on the local compute node without external API relay.

4. Strict LLM Boundary (No KYC to Gemini)
Google Gemini is isolated solely to conversational intent parsing on the advisor screen. **Zero document bytes, scanned images, tax identifiers, or extracted KYC personal records are ever transmitted to Gemini or external LLMs.**

5. File Validation & Path Traversal Guards
Strict MIME-type validation permits only PNG, JPEG, and PDF formats with a 10MB size ceiling. Storage paths are sanitised using UUID hashes, completely neutralizing directory traversal vulnerabilities (e.g. ../ attacks).

6. Sensitive Identifier Masking
Extracted 15-character GSTINs and 19-character Udyam numbers are masked in client-facing displays (e.g., 29*****1234*1Z5), safeguarding business identifiers against shoulder-surfing and unauthorized exposure.

7. Synthetic Registry Boundary & Limitations
All registry checks query local offline mock datasets (mock_gstin_registry.csv and mock_udyam_registry.csv). The prototype connects to no live government GSTN or Ministry portals; validity statuses reflect mock dataset fixtures.

8. Temporary Local Storage Architecture
Uploaded document artifacts are persisted strictly in local temporary application storage (local_uploads/). The prototype does not maintain permanent S3 or Blob vaulting, ensuring uploaded files are easily purged.

9. Prototype Authentication Limitations
Administrative analytics endpoints are gated via prototype HTTP header guards (X-Admin-Role: ExecutiveAdmin). Production readiness mandates migration to enterprise OAuth2 / Airtel Active Directory SSO.

10. Data Retention & Purge Boundaries
Onboarding sessions, orders, and follow-up notes reside in in-memory state stores. Session termination or server restart initiates full memory purge. Formal enterprise retention schedules remain subject to production data governance.

11. Synthetic Hackathon Dataset Disclosure
All lead telemetry (smb_leads.csv), customer profiles, and test accounts are synthetic datasets provided for evaluation. Zero real-world consumer PII is stored, ingested, or processed.

STATUTORY DISCLOSURE & LEGAL STATUS:

This prototype implements foundational technical safeguards (local OCR, privacy sandboxing, file sanitization, identifier masking, and consent gating). However, ***this hackathon prototype is NOT certified as fully compliant with the Digital Personal Data Protection (DPDP) Act 2023 or telecommunications KYC regulations.*** *Full commercial deployment requires independent legal privacy impact assessments, HSM-backed encrypted storage, and authorized UIDAI/GSTN API integration.*